[Demo] Seismic Evidence for a Subsurface Model

BANDAL Demo Workspace | Sample research note | 2026

Status: authored sample, not a real publication. Reflection times and velocities are invented. This is an arithmetic illustration, not an interpretation of real seismic data.

Abstract

This authored sample converts an invented two-way reflection time of 0.90 seconds to depth under three constant-velocity assumptions. Velocities of 2,200, 2,400 and 2,600 meters per second produce depths of 990, 1,080 and 1,170 meters. The exercise isolates model sensitivity and encourages readers to preserve assumptions alongside a reported estimate.

1. Constant-velocity example

For this simplified vertical-ray example, z = vt / 2. The factor of two accounts for the stated two-way time. We fix t = 0.90 seconds and vary only v. No dipping reflector, lateral variation or processing uncertainty is modeled.

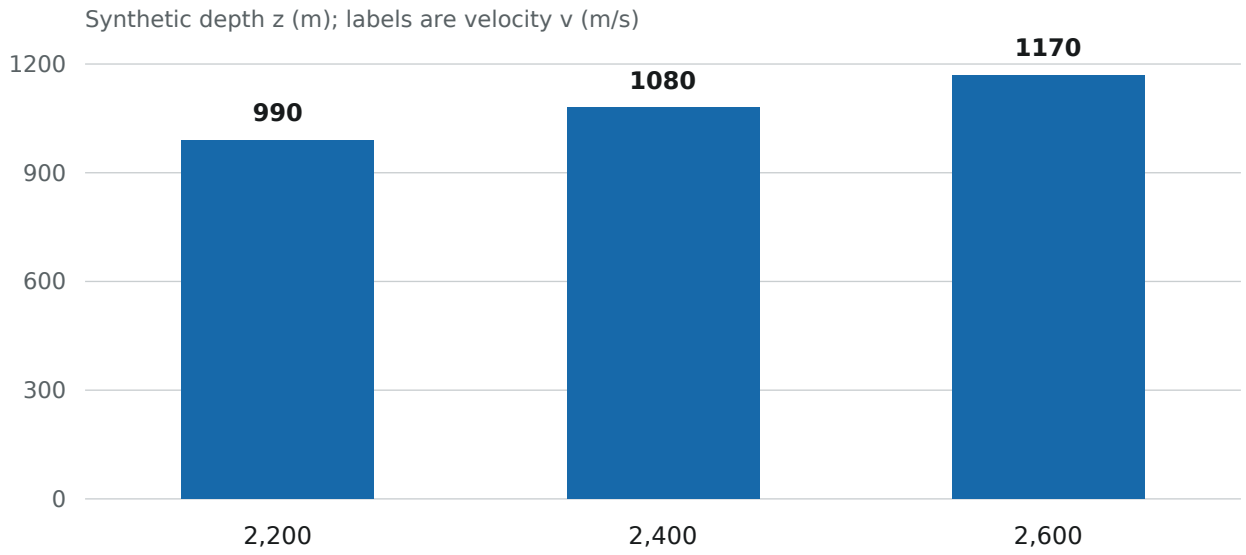


Figure 1. Invented depth estimates under three assumed velocities. The chart is a sensitivity exercise using synthetic values.

2. Synthetic time-to-depth inputs

Assumed velocity	Two-way time	Depth z	Offset from middle
2,200 m/s	0.90 s	990 m	-90 m
2,400 m/s	0.90 s	1,080 m	0 m
2,600 m/s	0.90 s	1,170 m	+90 m

Worked middle case: z = 2,400 x 0.90 / 2 = 1,080 meters. The three assumed velocities produce a 180-meter spread. The +/-90-meter range is a synthetic scenario range around the middle case, not a confidence interval.

3. Sensitivity and interpretation

At fixed time, the depth change is proportional to the velocity change: dz/dv = t/2 = 0.45 seconds. Raising the assumed velocity by 200 meters per second raises the estimated depth by 90 meters in this example. This statement follows the simplified model; it does not quantify real subsurface uncertainty.

4. Suggested reading notes

Question: Which velocity evidence would narrow the scenario range?
Claim to annotate: a scenario range is not a confidence interval.
Connection: compare model assumptions here with scoring assumptions in the CO2 storage screening note.

Provenance

Created specifically as a sample PDF for the BANDAL-hosted BluePrint demonstration. All data are synthetic and all calculations are transparent. No survey, external publication, DOI or scientific validation is claimed.

Reader exercise: select the time-to-depth formula and save a note that keeps the time, velocity assumption and resulting depth together.